

Stellenbosch University

Department of Industrial Engineering

Non-linear Optimisation 414

Tutorial 2: 6 March 2026

- (1) Which point on the plane $x_1 + x_2 + x_3 = 2$ in $\mathbb{R}^3$ is closest to the origin? What is this shortest distance? Motivate your reasoning and show all your numerical work.
- (2) Find all stationary points of the function $f(x_1, x_2) = 6x_1^2x_2 - 3x_1^3 + 2x_2^3 - 150x_2$ and classify these points as local minima, local maxima, or saddle points. Motivate your reasoning and show all your numerical work.
- (3) Perform two iterations of the method of gradient ascent to approximate the local maximum of the function

$$g(x_1, x_2) = 2x_1x_2 + 2x_2 - x_1^2 - 2x_2^2,$$

starting at the point $\mathbf{x}^0 = [\frac{1}{2} \quad \frac{1}{4}]^T$. Implement a backtracking Armijo-Goldstein line search with parameters $a = 0.45$ and $b = 0.8$ in Step 4 of Algorithm 3.2. Work accurately to three decimal places throughout and produce a table of algorithmic output similar in format to Table 3.1.

- (4) Perform two iterations of Newton's method for multivariate function optimisation (Algorithm 3.5) to approximate the local maximum of the function

$$h(x_1, x_2) = 4x_1x_2 - 2x_1^2 - x_2^4$$

in the rectangle $[-2, 0] \times [-2, 0]$, starting out from the initial estimate $\mathbf{x}^0 = [-2 \quad -\frac{3}{2}]^T$. Work accurately to three decimal places throughout and produce a table of algorithmic output similar in format to Table 3.5.